3.4.1 Integral Test Worksheet

Determine whether each of the following series converges or diverges. *<u>You may use any theorem from 3.4.1 or earlier.</u>*

1. $\sum_{k=1}^{\infty} \frac{1}{2k-1}$

2. $\sum_{k=1}^{\infty} \frac{3^k}{14k^2}$

3. $\sum_{n=1}^{\infty} \frac{2\sqrt{n}}{n^3}$

4. $\displaystyle\sum_{k=1}^{\infty}\frac{(-1)^k\cdot 2^{k+1}\cdot 3^k}{7^{2k}}$

5. $\displaystyle\sum_{n=1}^{\infty}\frac{3n+2}{n(n+1)}$ (*Hint: Use partial fractions to rewrite the function associated with this series*)

6. $\displaystyle\sum_{n=1}^{\infty}\frac{3n+2}{n+1}$

7. $\sum_{n=1}^{\infty} \frac{n^2}{n^3+1}$